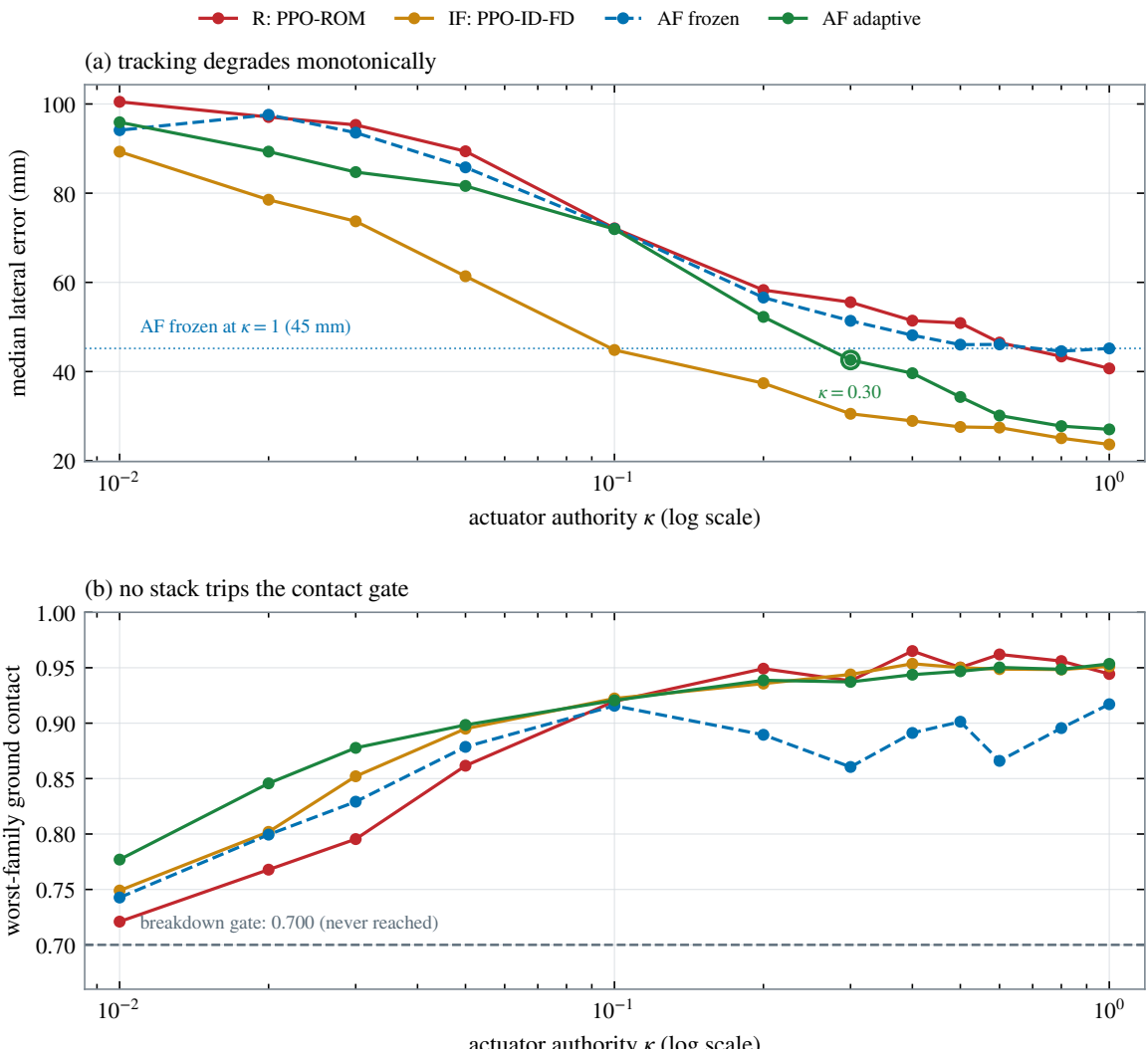
R: PPO-ROM
IF: PPO-ID-FD
AF frozen
AF adaptive
(a) tracking degrades monotonically
median lateral error (mm)
100
80
60
40
20
AF frozen at κ = 1 (45 mm)
κ = 0.30
10^−2
10^−1
10^0
actuator authority κ (log scale)
(b) no stack trips the contact gate
worst-family ground contact
1.00
0.95
0.90
0.85
0.80
0.75
0.70
breakdown gate: 0.700 (never reached)
10^−2
10^−1
10^0
actuator authority κ (log scale)